# Capstone Mission Defense

Prepare and defend the rover design using traceable requirements, test evidence, and risk-based recommendations.

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| **Advanced** | **2-3 class periods** | **Competition** |

## Engineering Context

A client review board must decide whether the rover is ready for mission demonstration. Teams will present concise evidence and respond to technical questions.

## What You Will Practice

- Synthesize design and test evidence
- Communicate technical tradeoffs
- Respond to review questions
- Make a defensible readiness recommendation

## What You Need

- Rover or capstone system
- Requirements and verification evidence
- Presentation software
- Engineering notebook
- Design review checklist

## Student Instructions

1. Select the five mission requirements most important to readiness.
2. Create one slide or visual for each requirement showing design response and verification evidence.
3. Include system architecture, key tradeoffs, major revision, and unresolved risk.
4. Prepare a five-minute presentation with assigned team roles.
5. Conduct a rehearsal and remove unsupported claims.
6. Present to the review board.
7. Answer questions using data, drawings, code, or test evidence.
8. Submit a final readiness recommendation and action list.

## Engineering Requirements

- Presentation is no longer than five minutes before questions
- Every major claim links to evidence
- Include at least one failed test or revision
- State unresolved risks honestly
- All team members contribute
- Final recommendation is Ready, Conditionally Ready, or Not Ready

## Constraints & Safety

- Use professional and respectful review behavior
- Do not conceal failed tests
- Follow safe demonstration boundaries
- No last-minute unverified hardware changes

## What You Submit

- Final presentation
- Requirement-to-evidence summary
- Risk register
- Board question log
- Readiness recommendation
- Individual reflection

## Reflection Questions

1. Which evidence was most persuasive?
2. What question exposed the weakest part of the design?
3. How did the team communicate uncertainty?
4. What action should happen before another mission attempt?

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| **Engineering Tip**<br>Lead with mission requirements and evidence, not a chronological story of everything the team did. | **Common Mistake**<br>Using confident language without test support can reduce reviewer trust. |
| **Stretch Goal**<br>Complete a second defense after addressing the review board's top three action items. | **Career Connection**<br>Program managers, chief engineers, and design-review boards decide whether aerospace systems are ready to proceed to the next phase. |